1.2 Schedule of activities

Table 1-1 Schedule of activities table

	Protocol section	Screening			In-house Baseline				Dose escalation/Dose maintenance							Dose maintenance					In-house End of treatment		In-house Follow-up			Ambulatory Follow-up			End of study	Withdrawal	
Visit		V1A[a]	V1B	V1C CGM	V2 D-3	V2 D-2	V2 D-1	V2D1 First dose	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14[p] CGM	V15 D-1	V15D1 Last dose	V15 D2	V15 D3	V15 D4	V16	V17	V18	V19	V19 X[q]	V19 Y
Timing of Visit / Study Day		-56	-17	-10	-3	-2	-1	1	15	29	57	85	113	141	169	197	225	253	263	274	280	281	282	283	284	288	295	302	308	-	≥ 28d after last dose
Timing of Visit / Study Week		-9	-3	-2	-1	-1	-1	0	2	4	8	12	16	20	24	28	32	36	37	39	40	40	40	40	40	41	42	43	44	-	-
Visit Window (Days)		+38	+7	+1	±1				±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	+7	±1	±1					±2	±3	±3	+3	-	-
Informed Consent[b]	10.1	X																													
Demography[c]	8.1	X																													
Eligibility Criteria[d]	5.1 5.2	X			X																										
Randomisation Criteria and Randomisation	5.4							X[e]																							
Discontinuation Criteria	7.1								X	X	X	X	X	X	X	X	X	X		X	X	X									
Medical History / Concomitant Illness[f]	8.1	X		X	X	X	X																								
Concomitant Medication	6.8	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X
Tobacco, E-Cigarette and Nicotine Products Use[g]	5.3.2	X			X																										
Screen For Alcohol[g]	10.2	X		X	X																X										
Screen For Drugs[g]	10.2	X		X	X																										
Childbearing Potential	10.4	X																													

	Protocol section	Screening			In-house Baseline				Dose escalation/Dose maintenance							Dose maintenance					In-house End of treatment		In-house Follow-up			Ambulatory Follow-up			End of study	Withdrawal	
Visit		V1A[a]	V1B	V1C CGM	V2 D-3	V2 D-2	V2 D-1	V2D1 First dose	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14[p] CGM	V15 D-1	V15D1 Last dose	V15 D2	V15 D3	V15 D4	V16	V17	V18	V19	V19 X[q]	V19 Y
Timing of Visit / Study Day		-56	-17	-10	-3	-2	-1	1	15	29	57	85	113	141	169	197	225	253	263	274	280	281	282	283	284	288	295	302	308	-	≥ 28d after last dose
Timing of Visit / Study Week		-9	-3	-2	-1	-1	-1	0	2	4	8	12	16	20	24	28	32	36	37	39	40	40	40	40	40	41	42	43	44	-	-
Visit Window (Days)		+38	+7	+1	±1				±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	+7	±1	±1					±2	±3	±3	+3	-	-
Contraceptive Counselling[h]	10.4	X			X					X	X	X	X	X	X	X	X	X			X									X	
Pregnancy Test[h,i]	8.3.8	X			X			X[e]		X[e]	X[e]	X[e]	X[e]	X[e]	X[e]	X[e]	X[e]	X[e]			X						X		X	X	X
Vital Signs	8.3.3	X		X	X			X[e]		X[e]	X[e]	X[e]	X[e]	X[e]	X[e]	X[e]	X[e]	X[e]			X						X		X	X	X
Clinical Outcome Assessments	8.2.7	X			X	X															X		X						X		X
C-SSRS Baseline		X																													
C-SSRS Since Last Visit					X																								X		X
Patient Health Questionnaire (PHQ-9)		X			X																								X		X
CoEQ 19 NRS					X																X										
VAS Appetite						X																	X								
VAS Appeal						X																	X								
Physical Examination	8.3.2	X																											X		X
Eye Examination[j]	8.3.4	X																				X								X	
ECG	8.3.5	X			X							X										X								X	
Adverse Event[f]	8.4						X	X	X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X
Hypoglycaemic Episodes	10.7							X	X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X

	Protocol section	Screening			In-house Baseline				Dose escalation/Dose maintenance							Dose maintenance					In-house End of treatment		In-house Follow-up			Ambulatory Follow-up			End of study	Withdrawal	
Visit		V1A[a]	V1B	V1C CGM	V2 D-3	V2 D-2	V2 D-1	V2D1 First dose	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14[p] CGM	V15 D-1	V15D1 Last dose	V15 D2	V15 D3	V15 D4	V16	V17	V18	V19	V19 X[q]	V19 Y
Timing of Visit / Study Day		-56	-17	-10	-3	-2	-1	1	15	29	57	85	113	141	169	197	225	253	263	274	280	281	282	283	284	288	295	302	308	-	≥ 28d after last dose
Timing of Visit / Study Week		-9	-3	-2	-1	-1	-1	0	2	4	8	12	16	20	24	28	32	36	37	39	40	40	40	40	40	41	42	43	44	-	-
Visit Window (Days)		+38	+7	+1	±1				±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	+7	±1	±1					±2	±3	±3	+3	-	-
Laboratory Assessments	10.2	X				X	X	X			X		X				X					X	X	X					X	X	X
Antibodies	8.9							X[e]									X[e]												X		X
PK Sampling	8.5							X[e]		X[e]	X[e]	X[e]	X[e]	X[e]	X[e]	X[e]	X[e]					X	X	X	X	X	X	X	X	X	X
Body Measurements	8.2.2	X	X			X				X	X	X	X	X	X	X	X	X	X			X							X	X	X
MRI Scan[k]	8.2.5		X																X												
Waist Circumference	8.2.2					X				X	X	X	X	X	X	X	X	X				X							X	X	X
Body Weight	8.2.2	X				X				X	X	X	X	X	X	X	X	X				X							X	X	X
Height	8.2.2	X																													
BMI	8.2.2	X				X				X	X	X	X	X	X	X	X	X				X							X	X	X
CGM Fitted	8.2.6			X																X											
CGM Removed and Data Uploaded	8.2.6							X[e]																	X						
Hand out and Instruct in BG meter	8.2.6							X																							
Collect BG meter	8.2.6																												X		X
Mixed Meal Tolerance Test	8.2.4					X																	X								

	Protocol section	Screening			In-house Baseline				Dose escalation/Dose maintenance							Dose maintenance					In-house End of treatment		In-house Follow-up			Ambulatory Follow-up			End of study	Withdrawal	
Visit		V1A[a]	V1B	V1C CGM	V2 D-3	V2 D-2	V2 D-1	V2D1 First dose	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14[p] CGM	V15 D-1	V15D1 Last dose	V15 D2	V15 D3	V15 D4	V16	V17	V18	V19	V19 X[q]	V19 Y
Timing of Visit / Study Day		-56	-17	-10	-3	-2	-1	1	15	29	57	85	113	141	169	197	225	253	263	274	280	281	282	283	284	288	295	302	308	-	≥ 28d after last dose
Timing of Visit / Study Week		-9	-3	-2	-1	-1	-1	0	2	4	8	12	16	20	24	28	32	36	37	39	40	40	40	40	40	41	42	43	44	-	-
Visit Window (Days)		+38	+7	+1	±1				±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	+7	±1	±1					±2	±3	±3	+3	-	-
Ad Libitum Meal[l]	8.2.4					X																	X								
Hyperinsulinaemic Euglycaemic Clamp	8.2.3.1						X																	X							
Hyperglycaemic Clamp with Arginine Stimulation Test[m]	8.2.3.2						X																	X							
Administration of Trial Product[n]	6.1							X	X	X	X	X	X	X	X	X	X	X		X		X[o]									
Dispensing Visit	6.1							X		X	X	X	X	X	X	X	X	X													
Drug Accountability[g]	6.2									X	X	X	X	X	X	X	X	X				X									
Overnight Fasting Required	5.3.1					X	X										X					X	X	X					X		X
Hand Out ID Card	8				X																										
Training in Trial Product, Pen-Handling	6.1.1							X		X																					
Hand Out Direction for Use	8							X																							
Hand Out and Instruct in Diary	8			X				X	X	X	X	X	X	X	X	X	X	X		X					X	X	X	X			
Collect and Review Diary	10.1.8				X				X	X	X	X	X	X	X	X	X	X		X	X					X	X	X	X	X	X

[a] For WOCBP at least 16 days are required between pregnancy test at screening and pregnancy test prior to dosing on V2D1.
[b] Informed consent must be obtained before any other study activities and assessments. The date and time of the informed consent must be recorded in the eCRF at screening on V1A.
[c] Demography consists of year of birth, sex (at birth) and race (according to local regulation). Race and ethnicity must be self-reported by the participant.
[d] Eligibility is assessed at V1A. Confirmation that the participant is eligible based on screening results is performed at V2D-3.
[e] Prior to dosing or assessments.
[f] Report conditions identified before first clamp procedure as medical history/concomitant illness in accordance with Section 8.1. Report conditions identified after first clamp procedure as AEs in accordance with Section 8.4.
[g] Results will not be recorded in the eCRF. Results to be kept in source data.
[h] Only applicable for WOCBP.
[i] Only pregnancy test at V1A is assessed in serum. Subsequent pregnancy tests are assessed in urine. In addition to the planned assessments, urine pregnancy tests should be performed at any time during the study if a menstrual period is missed, or if pregnancy is otherwise suspected.
[j] Eye examinations performed within 90 days prior to V1A and 21 days prior to V15D1 are also accepted. Eye examinations scheduled on V19X is only applicable for participants who withdraw/discontinue prior to V15; withdrawals/discontinuation after V15 are not required unless suspicion of diabetic retinopathy. Results from eye examinations scheduled on V19X do not need to be available at the same visit.
[k] MRI scan on V1B and V13 is to be performed within 7 days prior to CGM fitting. Deviation from this visit and/or assessment window will not be considered a protocol deviation.
[l] Served after the mixed meal tolerance test.
[m] Performed after the hyperinsulinaemic euglycaemic clamp. The hyperglycaemic clamp will end with the arginine test.
[n] QW self-administration at home except for the following visits: V2, V4, V5, V6, V7, V8, V9, V10, V11, V12 and V15, where trial product will be self-administered on-site under staff supervision.
[o] Last dose on V15D1 will be self-administered on site without drug dispensing.
[p] To remind participants on dosing and restrictions prior to in-house visit.
[q] Visit 19X should be performed as soon as possible after discontinuation or withdrawal if the participant has initiated study treatment.

Abbreviations: AEs = adverse events; BG = blood glucose; CGM = continuous glucose monitoring; CoEQ = control of eating questionnaire; C-SSRS = Columbia Suicide Severity Rating Scale; D = day; ECG = electrocardiogram; OAD = oral anti-diabetic drug; NRS = numerical rating scale; PK = pharmacokinetics; V = visit; VAS = visual analogue scale; WOCBP = women of childbearing potential.

Table 1-2 Overview of assessments including clinical outcome assessments, meal test, clamp procedures and pharmacokinetic blood sampling schedule

Intervention	Visit	Nominal time		PK sampling		Other assessments	COA
		Day	Hours[a]	NNC0487-0111	Window allowance[b]		
MRI scan[f]	V1B	-17	-	-	-	See Section 8	
CGM fitting	V1C	-10	-	-	-	See Section 8.2.6.	
In-house Baseline	V2D-3	-3	-	-	-		CoEQ
	V2D-2	-2	-	-	-	MMTT and *ad lib* meal. See Table 1-3 and Section 8. HEC and HGC. See Table 1-4 and Table 1-5 respectively.	VAS Appetite VAS Appeal
	V2D-1	-1	-	-	-		
Dose escalation	V2D1	1	00 (pre-dose)[c]	X	-30 min		
	V3	15	-	-	-		
	V4	29	00 (pre-dose)[d]	X			
	V5	57	00 (pre-dose)[d]	X			
	V6	85	00 (pre-dose)[d]	X			
	V7	113	00 (pre-dose)[d]	X			
	V8	141	00 (pre-dose)[d]	X			
	V9	169	00 (pre-dose)[d]	X			
Dose maintenance	V10	197	00 (pre-dose)[d]	X			
	V11	225	00 (pre-dose)[d]	X			
	V12	253	-	-	-		
Dose maintenance / MRI scan[f]	V13	263	-	-	-	See Section 8	
Dose maintenance / CGM fitting	V14	274	-	-	-	See Section 8.2.6.	
In-house End of treatment	V15D-1	280	-	-	-		CoEQ
	V15D1	281	00 (pre-dose)[c]	X	-30 min		
			02	X	±10 min		
			04	X	±10 min		
			08	X	±10 min		
			12	X	±10 min		
In-house Follow-up	V15D2	282	24[f]	X	±30 min	MMTT and *ad lib* meal. See Table 1-3 and Section 8. HEC and HGC. See Table 1-4 and Table 1-5 respectively.	VAS Appetite VAS Appeal
	V15D3	283	48[f]	X	±30 min		
	V15D4	284	72	X	±30 min		
Follow-up	V16	288	168	X	±2 days		
	V17	295	336	X	±3 days		
	V18	302	504	X	±3 days		
End of study	V19	308	648	X	+3 days		
Withdrawal[g]	V19X[h]	-	-	X	-		
	V19Y	≥ 28d after last dose	-	X	-		

[a] Indicates hours since dosing.

[b] Relative to dosing time.

[c] Blood to be sampled prior to first dosing (on V2D1 / Day 1) and prior to last dose (V15D1 / Day 281).

[d] Blood to be sampled prior to the next dose (applies to both dose escalation and dose maintenance).

[e] Blood to be sampled prior to MMTT (V15D2 / Day 282) and HEC and HGC (V15D3 / Day 283).

[f] MRI scan on V1B and V13 are to be performed within 7 days prior to CGM fitting. Deviation from this assessment window will not be considered a protocol deviation.

[g] Applicable for withdrawn or discontinued participants only. While there are no nominal time, sampling date and time to be recorded in the eCRF.

[h] Sampled as soon as possible after the last trial product administration.

Abbreviations: D = day; CoEQ = control of eating questionnaire; CGM = continuous glucose monitor; HEC = hyperinsulinaemic euglycaemic clamp; HGC = hyperglycaemic clamp; MMTT = mixed meal tolerance test; PK = pharmacokinetics; V = visit.

Table 1-3 Mixed Meal Tolerance Test blood sampling and visual analogue scale assessment schedule at V2D-2 and V15D2

					Blood sampling									
Nominal time (min)	**Window allowance (min)**	**Meal**	**VAS Appetite**	**VAS Appeal**	**Glucose**	**Insulin, C-peptide**	**Proinsulin, Soluble Leptin Receptor, Leptin**	**Glucagon**	**PYY 3-36**	**Amylin (endogenous)**	**ApoB48, Adiponectin (total and HMW)**	**FFA**	**TG, VLDL-C**	**TC, HDL-C, LDL-C**
-15			X	X										
-10	±5				X	X								
-1	-4				X	X	X	X	X	X	X	X	X	X
0		Start												
15	+10[a]	Stop	X	X	X	X		X		X				
30	+5		X		X	X		X	X	X		X		
60	±5		X		X	X		X	X	X	X	X	X	
90	±5		X		X	X		X		X		X	X	
120	±5		X		X	X			X		X	X	X	
150	±5				X	X								
180	±10		X		X	X		X	X	X	X	X	X	
240	±10		X		X	X		X			X	X	X	
300	±5		X		X	X		X	X	X	X	X	X	

[a] Mixed meal should be consumed within 15 minutes. A maximum of 25 minutes is allowed in case of severe difficulties. Meal start and end times should be recorded in the eCRF.

Abbreviations: ApoB48 = apolipoprotein B48; FFA = free fatty acids; HDL-C = high-density lipoprotein cholesterol; HMW = high molecular weight; LDL-C = low-density lipoprotein cholesterol; MMTT = mixed meal tolerance test; PYY 3-36 = peptide YY 3-36; TC = total cholesterol; TG = triglycerides; VAS = visual analogue scale; VLDL-C = very low-density lipoprotein cholesterol.

4 Study design

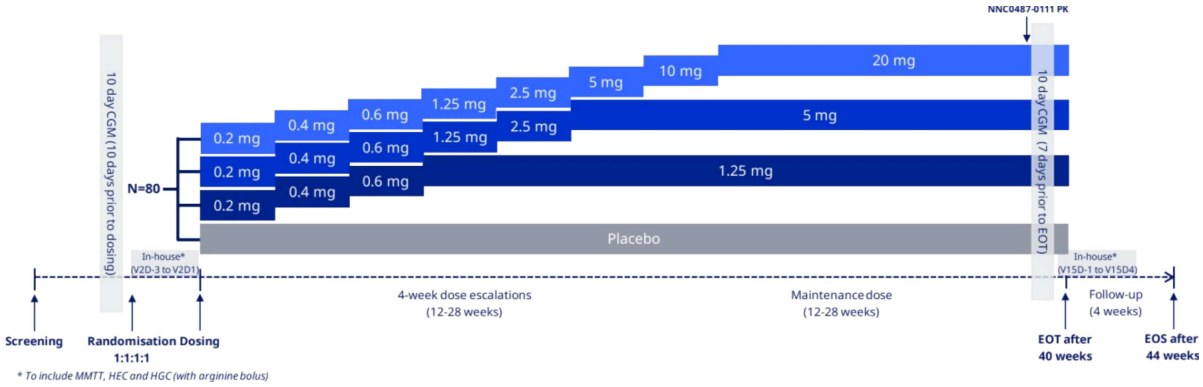
4.1 Overall design

This is an interventional, single-centre, randomised, placebo-controlled, double-blind, multiple-dose study. A total of 80 study participants with T2D will be randomised in a 1:1:1:1 ratio to QW NNC0487-XXXX 20 mg, 5 mg, 1.25 mg or placebo, respectively, stratified according to sex (male/female) and HbA_{1c} at screening (<7.0% / ≥7.0%).

The study consists of:

- an up to 52-day screening period.
- a 4-day (3 nights) in-house visit with baseline assessments; body measurements, MMTT, HEC, and HGC being performed before first dosing (pre-treatment).
- a 40-week treatment period (with dose escalation) on NNC0487-XXXX 20 mg, 5 mg, 1.25 mg, or placebo.
- end of treatment will include a 5-day (4 nights) in-house visit with MMTT, HEC and HGC performed.
- a minimum 4-week safety follow-up period.

Figure 4-1 Study design



Abbreviations: CGM = continuous glucose monitoring; D = day; EOS = end of study; EOT = end of treatment; HEC = hyperinsulinaemic euglycaemic clamp; HGC = hyperglycaemic clamp; MMTT = mixed meal tolerance test; V = visit.

Treatment with NNC0487-XXXX and placebo will be initiated at 0.2 mg or placebo QW, followed by stepwise fixed dose escalation every 4 weeks as shown in Figure 4-1. After reaching the target dose of NNC0487-XXXX 20 mg, 5 mg or 1.25 mg, this dose level will be maintained for up to 28 weeks, depending on dose. A total of 4 doses of investigational medicinal product (IMP) will be administered at each dosing step before escalating to a higher dose.

Participants will attend the site for 21 visits. Details of other assessments and reminders during different visits can be found in the schedule of activities (SoA, Table 1-1). Following the screening visit (V1A), participants will return for body composition assessments (via magnetic resonance imaging (MRI), V1B) and for a continuous glucose monitoring (CGM)-fitting visit (V1C), before attending an in-house visit (V2). During the baseline visit, a MMTT, a HEC and a HGC will be performed. Participants will have their CGM removed after wearing it continuously for 10 days,

prior to being randomised and receiving the first IMP dose before leaving the site. Subsequently, participants will self-administer the IMP once weekly at home (except V2, V4, V5, V6, V7, V8, V9, V10, V11, V12 and V15, where trial product will be administered on site). The study period includes 40 weeks of treatment with dose escalation in the first 12-28 weeks and dose maintenance in the last 12-28 weeks of treatment. Participants will visit the site approximately every 4 weeks for dose escalation in the dose escalation period. Frequent PK sampling visits are scheduled from end of treatment and during follow-up period from V15 to V19. Participants will have another body composition assessed at V13 (MRI), and a CGM fitting at V14. Participants will attend the site for an end of treatment in-house visit (V15), where the last dose will be administered, a MMTT, a HEC and a HGC will be performed. There will be a 4-week follow-up period after end of treatment.

4.2 Scientific rationale for study design

The treatment duration of 40 weeks is considered adequate to ensure weight-loss-dependent effects on insulin sensitivity and pancreatic secretory function. Due to the length of the treatment (40 weeks) and follow-up period (4 weeks), a parallel-group design has been selected.

A total of 80 participants will be randomised in a 1:1:1:1 ratio to NNC0487-XXXX 20 mg, 5 mg, 1.25 mg or placebo. Randomisation will be stratified by sex (male/female) and HbA_{1c} at screening (<7.0% / ≥7.0%). The aim is for at least 60% of randomised participants to have an HbA_{1c} ≥7.0% at screening. This will be monitored throughout the recruitment period, and the Sponsor may implement supportive strategies as needed to help maintain this threshold. Inclusion of the placebo arm will enable characterisation of the mechanism of action induced by NNC0487-XXXX.

Patients with overweight or obesity with/without T2D are the target population for the current clinical development programme for NNC0487-XXXX. Patients with T2D have dysfunctional glucose metabolism, thus, the impact of treatment with NNC0487-XXXX on insulin sensitivity and pancreatic endocrine function is most conveniently studied in the target population. Participants on treatment with metformin with or without sodium-glucose cotransporter-2 inhibitors (SGLT2)i are selected, as they are expected to be in the earlier stages of T2D with relatively preserved β-cell function, and hence appropriate for investigating the mechanism of action of NNC0487-XXXX.

Participants should be on stable metformin treatment during the study as this remains a foundational therapy in the management of T2D. The inclusion of participants with excess body weight (BMI 27.0 – 45.0 kg/m^2) is due to the expected weight loss in response to 40 weeks of treatment with NNC0487-XXXX. A wide age range (18-75 years) is included to reflect the target population for the treatment, albeit with an upper age limit of 75 years to reduce the number of comorbidities in the study population.

MRI-based body composition assessment is included in this study to provide a comprehensive, non-invasive evaluation of treatment effects on fat distribution, lean mass preservation and bone parameters. MRI enables quantification of visceral and subcutaneous adipose tissue and muscle quality, as well as organ-specific fat content and bone characteristics, offering insights into the metabolic quality of weight loss. The inclusion of MRI supports mechanistic understanding of NNC0487-XXXX.

The HEC method is chosen because it is considered the gold standard for the evaluation of insulin sensitivity (M-value).[13] During HEC, a constant insulin infusion rate (IIR) will be used to ensure

that the glucose infusion rate (GIR) at steady state conditions of euglycaemia (5.5 mmol/L or 100 mg/dL) is equal to glucose uptake in all tissues in the body (i.e. whole-body insulin sensitivity). The HGC will be used to evaluate β-cell function via an acute raise of blood glucose (BG) concentration to a predefined hyperglycaemic plateau (12 mmol/L or 216 mg/dL), where endogenous insulin secretion (first and second phase insulin secretion, via C-peptide assessment and de-convolution methodology) and other parameters of interest is measured. An arginine bolus will be given at the end of the HGC (120 min) to stimulate insulin secretion, enabling evaluation of insulin secretory response after a period of steady state hyperglycaemia. The MMTT will be used to investigate pancreatic hormone secretion and parameters related to postprandial metabolism mimicking normal physiological conditions after intake of a meal.

M-value is chosen as the primary endpoint, as it represents a measure of whole-body insulin sensitivity. Supportive secondary endpoints are chosen based on the current understanding of the physiological and pharmacological effects of amylin and GLP-1, to support the understanding of mechanism of action for NNC0487-XXXX.

4.3 Justification for dose

It is well known that to mitigate gastrointestinal (GI) side effects with GLP-1 RA treatment, a gradual dose escalation to the target dose is required. Participants will be initiated at a QW dose of 0.2 mg and follow a fixed-dose escalation regimen, with dose increasement every 4 weeks, to target doses of 1.25 mg, 5 mg and 20 mg, reached after 12-28 weeks.

The starting dose, escalation regimen and maintenance doses of NNC0487-XXXX are selected based on the safety, tolerability, and pharmacokinetics (PK) observed in the phase 1b/2a study investigating the NNC0487-XXXX in a s.c. formulation (NN9490-7613). In study NN9490-7613, multiple ascending QW s.c. doses of NNC0487-XXXX were all deemed tolerable and safe up to 60 mg with the dose escalation being applied. As expected, a dose dependent increase in GI adverse events (AEs) was observed, as seen for other GLP-1 RAs, and has been shown for the long-acting amylin analogue cagrilintide alone and in combination with the GLP-1 RA semaglutide.

To further support GI tolerability in the current study, two lower doses (0.2 mg and 0.4 mg QW) have been introduced. These doses were not included in the previous NN9490-7613 study and are intended to enable a more gradual escalation, which is a well-established strategy for reducing GI side effects associated with GLP-1 RA therapy.

Further information about the expected exposure for NNC0487-XXXX in a s.c. formulation can be found in the current edition of the Investigator’s Brochure and any updates hereof.

4.4 End of study definition

The end of the study is defined as the date of the last scheduled visit of the last participant in the study.

A participant is considered to have completed the study if he/she has completed all periods of the study including the last visit. The primary endpoint is evaluated at V15. The primary completion date (PCD) is defined as the date on which the last participant in the clinical study has an assessment for the primary endpoint. If the last participant is discontinued early, the PCD is